

Question			Marking details		Marks available				
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Force is upward <u>on the magnet</u> (1) [F]LHR quoted and application of N3 (1) Force is down on the wire (1)	1	1 1		3		3
		(ii)	Mean calculated = 0.0236 or 0.02355 (1) accept 2.4×10^{-2} Absolute uncertainty calculated or implied (± 0.0012) (1) % uncertainty = 5% (1) accept 1 or 2 sig figs accept range 4 – 5.1%		3		3	3	3
		(iii)	Gradient = Bl used or implied (1) $B = 0.295$ [T] (no sig fig penalty) (1) Adding % uncertainties ecf from (a)(ii) (11%) (no sig fig penalty) (1) accept first principles approach (0.295 ± 0.032)T or (0.30 ± 0.03) ecf must have consistency of sig figs either 1 or 2 sig figs in uncertainty + UNIT or equivalent in mT (1)		4		4	4	4
	(b)	(i)	Force on charge carriers due to mag field or [F]LHR (1) Charge carriers go to one side of conductor / reference to build-up of charge (1) Electric field set up or top is +ve / -ve depending on carriers (1)	3			3		3
		(ii)	Indicative content: 1. Line goes through origin. 2. Graph is a straight line. 3. Line passes through all the error bars. 4. Both values of B compared. 5. Hall probe is more precise or its uncertainty is smaller [accept accurate]. 6. Sensible comment comparing methods e.g. method A is graphical, method B required only one measurement or converse, method A is more prone to human error, method B doesn't confirm $F = BIl$			6	6	3	6

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			5-6 marks 5 or 6 points clearly described <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks 3 or 4 points clearly described <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks 1 or 2 points clearly described <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
			Question 4 total	4	9	6	19	10	19